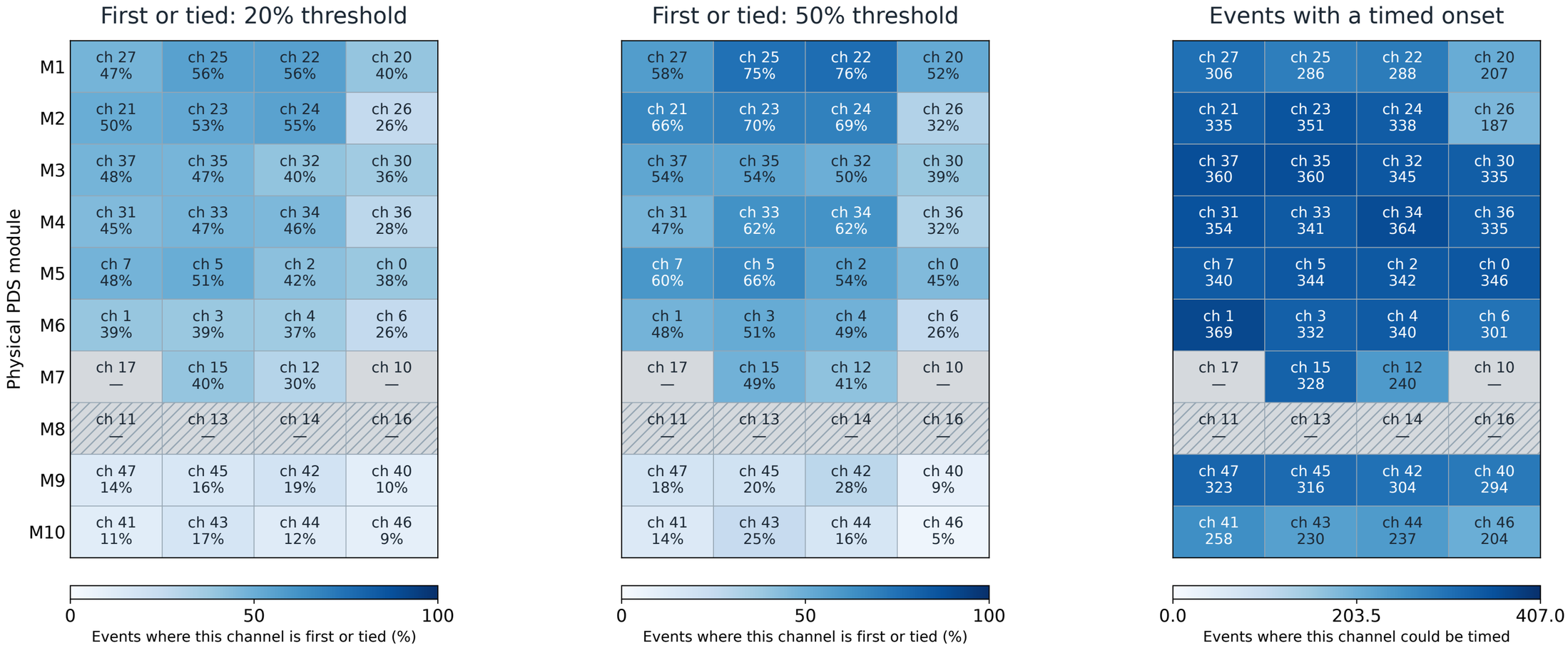
APA2: which channels tend to start first?

407 recorded pulses before the TPC precursor · Runs 32731 and 32739 · Native 16 ns samples

Each pulse needs at least eight connected channels with a recorded quiet baseline. Missing channels cannot be ranked.



At 20% pulse depth: 260 / 407 events have tied leaders. At 50%: 365 / 407 have tied leaders.

“First or tied” means within 16 ns of the earliest measured connected channel. Percentages use each channel’s own recorded sample.

Channels 22 and 25 are often early; no timed channel always leads. Pulse shape and electronics delays can affect the order.

Grey: no usable timing. Hatched: disconnected inputs, excluded from rankings. No correction for cable or electronics delays.

3-sample median filter; 32-sample baseline; thresholds also require ≥10 ADC counts and ≥5 baseline-noise units. One-bin ties are a convention, not calibrated precision.